

Carmen Cabrera

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Education

- 2017-21 PhD in Applied Mathematics, University College London (UK)
Thesis: Mathematical modelling of social phenomena in urban areas
Supervisor: Prof Steven Bishop
Examiners: Prof Denise Pumain (University Paris I Pantheon-Sorbonne) and Prof Frank Smith (UCL)
- 2016-17 MSc in Complex Systems, King's College London (UK)
Dissertation: Random matrices spectra, towards the solution of the Bray-Rodgers equation
Supervisor: Dr Pierpaolo Vivo
Grade: Distinction
- 2012-16 BSc in Physics, University of Valencia (Spain) with year abroad at Imperial College London (UK)
Dissertation: Obtaining geometric characteristics of a directed acyclic graph from its causal structure
Supervisor: Dr Tim Evans
Grade: 9.15 out of 10, equivalent to First Class Honours

Current academic appointments

- 2022- Lecturer in Geographic Data Science, Geographic Data Science Lab (GDSDL), University of Liverpool

Honorary appointments

- 2022- Honorary Lecturer, Centre for Advanced Spatial Analysis, UCL

Prior academic appointments

- 2021-22 Postdoctoral Research Fellow in Urban Mobility, Centre for Advanced Spatial Analysis, UCL
2017-21 Postgraduate Teaching Assistant, Department of Mathematics, UCL

Publications

PEER-REVIEWED JOURNAL ARTICLES

- 2026 A Renninger, C Cabrera. Extreme heat reduces and reshapes urban mobility. *PNAS Nexus* (in press). [arXiv:2501.03978](https://arxiv.org/abs/2501.03978).
- 2025 C Cabrera, R Neville. [Widely used but barely trusted: understanding student perceptions on the use of generative AI in higher education](#). *Perspectives: Policy and Practice in Higher Education* 1–14.
- 2025 M González-Leonardo, C Cabrera, R Neville, Andrea Nasuto, Francisco Rowe. [Beyond the Immediate Impacts of COVID-19 on Internal Population Movements in Mexico: Facebook Data Reveal Urban Decay and Slow Recovery—A Research Note](#). *Demography* 62(4): 1171–1184.

- 2025 [C Cabrera](#). [Spatial network analysis](#). *REGION*.
- 2025 A E Frazier, T Nelson, P Kedron, E Shook, S Dodge, A Murray, M Goodchild, S Battersby, J I Blanford, C Claramunt, J Holler, C Koylu, A Lee, S Manson, S Şalap-Ayça, J P Wilson, B Zhao, L Bennett, [C Cabrera](#), R Franklin, G McKenzie, H Miller, T Oshan, S Rey, F Rowe, S Spielman, W Xu. [Rethinking GIScience education in an age of disruptions](#). *Transactions in GIS*.
- 2024 F Rowe, [C Cabrera](#), M González-Leonardo, A Nasuto, R Neville. [Medium-term changes in the patterns of internal population movements in Latin American countries: effects of the COVID-19 pandemic](#). *UN-ECLAC: Population and development*.
- 2024 T Nelson, A E Frazier, P Kedron, S Dodge, B Zhao, M Goodchild, A Murray, S Battersby, L Bennett, J I Blanford, [C Cabrera](#), C Claramunt, R Franklin, J Holler, C Koylu, A Lee, S Manson, G McKenzie, H Miller, T Oshan, S Rey, F Rowe, S Şalap-Ayça, E Shook, S Spielman, W Xu, J Wilson. [A research agenda for GIScience in a time of disruptions](#). *International Journal of Geographic Information Science* 1–24.
- 2024 M González-Leonardo, [C Cabrera](#), R Neville, A Nasuto, F Rowe. [COVID-19 y movimientos de población entre la jerarquía urbana en México. Un análisis utilizando datos digitales - COVID-19 and internal population movements across the urban hierarchy in Mexico. An analysis using digital data](#). *Estudios Demográficos y Urbanos* 39, 3.
- 2023 [C Cabrera](#), C Zhong, M Batty, R Silva, SM Kang. [Inferring urban polycentricity from the variability in human mobility patterns](#). *Scientific Reports* 13.
- 2022 Y Wang, C Zhong, Q Gao, [C Cabrera](#). [Using Twitter to track internal migration before and during the COVID-19 pandemic in the UK](#). *Urban Informatics* 1, 15.
- 2022 R Prieto-Curiel, [C Cabrera](#), S R Bishop. [Scaling beyond cities](#). *Frontiers in Physics* 10.
- 2021 [C Cabrera](#), S R Bishop. [Urban population size and road traffic collisions in Europe](#). *PLOS ONE* 16(8): e0256485.
- 2020 [C Cabrera](#), S R Bishop. [The effect of dragon-kings on the estimation of scaling law parameters](#). *Scientific Reports* 10: 20226
- 2020 [C Cabrera](#), R Prieto-Curiel, S R Bishop. [Uncovering the behaviour of road accidents in urban areas](#). *Royal Society Open Science* 7: 191739
- 2020 R Prieto-Curiel, [C Cabrera](#), M Torres Pinedo, H González Ramírez, S R Bishop. [A continuous binning for discrete, sparse and concentrated observations](#). *Methods X* 7: 100709
- 2019 R Prieto-Curiel, [C Cabrera](#), M Torres Pinedo, H González Ramírez, S R Bishop. [Temporal and spatial analysis of the media spotlight](#). *Computers, Environment and Urban Systems* 75: 254 – 263

BOOK CHAPTERS

- 2025 R Prieto-Curiel, [C Cabrera](#). [Modeling Urban Population Dynamics and City-to-City Migration](#). *Compendium of Urban Complexity*.

PREPRINTS

- 2025 [C Cabrera](#), F Rowe. [A systematic machine learning approach to measure and assess biases in mobile phone population data](#). *arXiv:2509.02603*.
- 2025 [C Cabrera](#), M González-Leonardo, A Nasuto, R Neville, F Rowe. [Sustained changes to urban mobility after COVID-19 amplified socio-economic inequalities in Latin America](#). *arXiv: 2504.15871*.
- 2024 N Newsham, P Kumar, [C Cabrera](#), and F Rowe [A Long Short-Term Memory Forecast of sub-National Population Change Across Europe](#). *OSF preprints 10.31235/osf.io/9azqy*.
- 2024 M González-Leonardo, [C Cabrera](#), R. Neville, A. Nasuto, F. Rowe. [“Urban Exodus” During COVID-19 in Mexico? Using Digital Data to Analyze Medium-Term Pandemic Impacts on Internal Population Movements](#). *OSF preprints 10.31219/osf.io/e4au9*.

- 2023 [C Cabrera, LW Ng, H Wong, C Zhong. The temporal concentration of travel demand in an urban transport network. *arXiv:2310.02385v1*.](#)
- 2023 [A Renninger, V Marin Maureira, C Cabrera, R Prieto-Curiel. Detecting the sensitive spots in the African interurban transport network. *arXiv:2307.06402v1*.](#)

OTHER

- 2021 Alan Turing Institute Data Study Group team. [Data Study Group Final Report: Odin Vision. Zenodo](#)
The Alan Turing Institute, Publications & policy submissions.
- 2021 [C Cabrera. Mathematical modelling of social phenomena in urban areas. Doctoral thesis \(PhD\). University College London.](#)

Funding

RESEARCH GRANTS

Research income – total: £467,377.81 (USD 625,290.75)

- 2025-26 *Economic and Social Research Council (ESRC)* - “Building Imago-HASP collaborative infrastructure for joint data products linking satellite, synthetic and trajectory data”. CoI. 2024. £95,901.54.
- 2025 *Research England, University of Liverpool* - “Establishing a new cross-sector, collaborative agenda for data-driven transport planning in Liverpool City Region”. Partnership Innovation Fund, 185525. CoI. 2024. £22,342.00.
- 2024-26 *Economic and Social Research Council (ESRC)* - “DEBIAS. Developing a framework to measure and correct biases in human mobility data extracted from digital footprints”. CoI. 2024. £185,084.14.
- 2024 *Research England, University of Liverpool* - “TIMELY. Validating an up-to-date dataset of internal migration to inform local and regional-scale spatial planning strategies in Chile”. Public Support Fund, 182341. CoI. 2024. £74,136.53.
- 2023 *Research England, University of Liverpool* - “RECAST. Strengthening partnerships with the United Nations: Building capacity in digital footprint data for COVID-19 recovery”. Public Policy Quality-Related Pump-Priming Fund, 177990. CoI. 2023. £89,913.60.

TRAVEL GRANTS AND FUNDED RESEARCH VISITS

- 2024- *Center for Monetary and Financial Studies (CEMFI), Spain* - Research visit. Travel grant: 2,500€.
- 2024 *Barcelona Supercomputing Centre, Spain* - Research visit. Travel grant: 1,000€.
- 2023 *University of California Santa Barbara, US* - Specialist Meeting 2023. Travel grant: \$800.
- 2018 *NTU, Singapore* - Winter School on Complexity Science, Singapore. Travel grant: £1,837.
- 2015 *IFISC (CSIC-UIB), Spain* - “Chimera states in an oscillator network under non-local coupling effects”. Summer Undergraduate Research Fellowships. PI. 2015. 1,100€.

SCHOLARSHIPS AND STUDENTSHIPS

- 2017-21 *Department of Mathematics, UCL*. Teaching Assistantship to complete fully-funded PhD studies.
- 2019-20 *Department of Mathematics, UCL*. Monica Hulse Scholarship, awarded yearly to one postgraduate student based on academic merit. £500.
- 2016-17 *Department of Mathematics, King’s College London*. E T Davies Scholarship, awarded yearly to one or two outstanding applicants. £1,000.
- 2016-17 *La Caixa*. Scholarship awarded to 65 top Spanish candidates to complete postgraduate studies abroad. £27,000.

Honours and awards

- 2025 Learning, Teaching and Student Experience awards 2025/26, University of Liverpool
- 2024 Fellow of the Higher Education Academy
- 2021 Associate Fellow of the Higher Education Academy
- 2021 Nominated & invited to the Global Young Scientists Summit, National Research Foundation, Singapore

Talks

KEYNOTE TALKS

- 2026 The Oxford Summer School in Economic Networks 2026, University of Oxford, UK
- 2026 Winter Workshop on Complex Systems 2026, Mallorca, Spain
- 2025 16th International Conference on Complex Networks (Complenet) 2025, Fortaleza, Brazil
- 2024 Conference on Complex Systems (UrbanSys), Exeter, UK
- 2022 Early Career Researchers in Mathematics (ECRM) conference, UCL, UK

INVITED SEMINAR TALKS

- 2026 Institute of Environmental Science and Technology (ICTA), Universitat Autònoma de Barcelona, Spain
- 2025 School of Civil Engineering, University of Sydney, Australia
- 2024 BT, London, UK — online
- 2024 Spatial, Computational, and Real-Time Analytic Methods (SCRAM) Working Group — online
- 2024 Escuela de Negocios, Universidad Adolfo Ibáñez, Viña del Mar, Chile
- 2024 Departamento de Ciencias de la Computación, Universidad de Chile, Santiago, Chile
- 2024 Urban Beers, Santiago, Chile
- 2024 EchonophysiX, Ecole Polytechnique, Paris, France — online
- 2023 EdinbR, Edinburgh, UK
- 2023 Department of Human Geography lecture series, University of Toronto, ON, Canada — *Cancelled*
- 2023 Quantitative Spatial Science seminar, University of Bristol, UK
- 2022 Disordered Systems Group seminar, King's College London, UK
- 2022 Geographic Data Science Lab seminar, University of Liverpool, UK
- 2022 Centre for Complexity Science seminar, Imperial College London, UK
- 2022 Networks seminar, Centre for Advanced Spatial Analysis, UCL, UK
- 2021 Networks seminar, Centre for Advanced Spatial Analysis, London, UK
- 2021 Disordered Systems Group seminar, King's College London, UK
- 2020 Complexity Lates, London Institute for Mathematical Sciences, UK
- 2019 Postgraduate Seminar, Dept of Mathematics, UCL, UK

CONTRIBUTED TALKS

- 2025 British Society for Population Studies Annual Conference, Swansea, UK
- 2025 International Population Conference, Brisbane, Australia
- 2024 British Society for Population Studies Annual Conference, Bath, UK
- 2024 Complenet 2024, Exeter, UK — *Awarded best talk*
- 2023 Korean Research Institute of Human Settlements & GDSL knowledge exchange, Seoul, South Korea
- 2023 GISRUK, Glasgow, UK
- 2022 Korean Research Institute of Human Settlements & GDSL knowledge exchange, Liverpool, UK
- 2022 Conference on Complex Systems, IFISC, Palma de Mallorca, Spain
- 2022 GISRUK, Liverpool, UK
- 2020 Graduate Student Meeting, London Mathematical Society, UK — *Awarded best talk*
- 2019 Poster presentation at Latin American Conference on Complex Networks, Cartagena, Colombia

Conferences, workshops and events

ORGANISATION (SEE SERVICE SECTION FOR CONFERENCE ROLES)

- 2026 Project symposium “DEBIAS Symposium and Workshop: Setting standards for bias-adjusted digital mobility data”, Liverpool, UK
- 2026 Training workshop “Correcting Biases in Mobile-Phone Mobility Data: The DEBIAS Framework”, Mobile Tartu International Conference and PhD School, Tartu, Estonia
- 2024 Working group meeting 1 for Nowcasting Populations: “Understanding the opportunities, limitations and challenges of digital data sources for nowcasting populations”, University of Liverpool, Liverpool, UK
- 2024 Training workshop “An introduction to network analysis”, British Society for Population Studies Annual Conference, Bath, UK
- 2024 Expert workshop for the “Validation of a dataset on internal migration”, Ministry of Social Development, Santiago, Chile
- 2023 Training workshop “An Introduction to Network Analysis for GIS”, GIScience 2023, Leeds, UK
 - Workshop website: <https://carmen-cabrera.github.io/nets4gis/>
- 2023 Training Workshop “RECAST Project - Use of digital traces for the measurement and monitoring of population movements”, United Nations - ECLAC, Santiago, Chile
 - Workshop website: <https://fcorowe.github.io/dfd4mobility/>
- 2023 Policy Workshop “RECAST Project”, United Nations - ECLAC, Santiago, Chile
- 2023 Workshop “RECAST Project - Developing a Collaborative Project Plan”, United Nations - DPPA, New York City, NY, US
- 2023 Research Roadshow, School of Environmental Sciences, University of Liverpool, UK

CONTRIBUTION AND PARTICIPATION

- 2025 Workshop and Hackathon “Dynamic Estimates of Displacement in Disaster Regions: A Policy-Driven Framework Triangulating Data”, UN International Office for Migration, Berlin, Germany
- 2025 Cities as Complex Systems (CtCS2025) – Lessons Learned from Three Decades of Urban Complexity Science: Formulating Policy Recommendations, Complexity Science Hub, Vienna, Austria
- 2025 Working group meeting 4 for Nowcasting Populations: “Final session”, University of Oxford, Oxford, UK.
- 2025 Working group meeting 3 for Nowcasting Populations: “Methods session II”, University of Southampton, Southampton, UK.
- 2024 Working group meeting 2 for Nowcasting Populations: “Methods session I”, University of Manchester, Manchester, UK.
- 2024 Working group meeting 1 for Nowcasting Populations: “Understanding the opportunities, limitations and challenges of digital data sources for nowcasting populations”, University of Liverpool, Liverpool, UK.
- 2024 British Society for Population Studies Annual Conference, Bath, UK
- 2024 Conference on Complex Systems (UrbanSys), Exeter, UK
- 2024 AGILE 2024, Glasgow, UK
- 2024 CompleNet 2024, Exeter, UK
- 2023 Specialist Meetings 2023: Spatial Data Science in an Age of Scientific Disruption, University of California Santa Barbara, Santa Barbara, US
- 2023 GIScience 2023, Leeds, UK
- 2023 Korean Research Institute of Human Settlements & GDSDL knowledge exchange, Seoul, South Korea
- 2023 GISRU23, Glasgow, UK
- 2023 Turing-Newcastle scoping workshop on “Data and Data Science for the Measurement, Understanding and Mitigation of Place-Based Disparities”, Newcastle University, UK
- 2022 Workshop ‘UA 2.0 – Inequalities in the Recovery of Cities and Regions’, Alan Turing Institute Urban Analytics, Liverpool, UK
- 2022 Korean Research Institute of Human Settlements & GDSDL knowledge exchange, Liverpool, UK
- 2022 Conference on Complex Systems 2022, Palma de Mallorca, Spain

2022	GISRUK22, Liverpool, UK
2021	Global Young Scientists Summit, National Research Foundation, Singapore
2019	MacArthur Workshop on Urban Modelling & Complexity Science, CASA (UCL), London, UK
	Talking Maths in Public, Isaac Newton Institute, University of Cambridge, Cambridge, UK
2019	LANet2019 - Latin American Conference on Complex Networks, University of Los Andes, Cartagena, Colombia
2019	Workshop on Economic Science with Heterogeneous Interacting Agents, UCL and City University of London, UK
2019	Enrico Fermi Intl. School on Computational Social Science and Complex Systems, Varenna, Italy
2018	NTU Winter School 2018 on Complexity Science, Nanyang Technological University, Singapore

Teaching

POSTGRADUATE

University of Liverpool

2022-	<i>Module lead:</i> Computational Social Science - Module website: https://www.population-science.net/
2023-	<i>Lecturer:</i> Geographic Data Science - Module website: https://gds-l-ul.github.io/gds/
2022-	<i>Supervisor:</i> MSc in Geographic Data Science Dissertation (total of 3 students)

Centre for Monetary and Financial Studies (CEMFI)

2024-	<i>Lecturer:</i> Geographic Data Science for Applied Economists (part of the Urban Economics course taught by Prof. Diego Puga) - Module website: https://carmen-cabrera.github.io/gds4eco/
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University College London

2021-	<i>Supervisor:</i> MSc in Urban Spatial Science Dissertation at CASA (total of 7 students)
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UNDERGRADUATE

University of Liverpool

2025	<i>Lecturer:</i> Field Class Year 3 (BA in Geography) in Barcelona, Spain
2023-	<i>Supervisor:</i> BA in Geography Dissertation (15+ students)
2022-25	<i>Lecturer:</i> Field Class Year 3 (BA in Geography) in Toronto, ON, Canada
2022-24	<i>Lecturer:</i> Contemporary Population Dynamics
2022-23	<i>Lecturer:</i> Population and Societies

University College London

2018-20	<i>Module lead:</i> Mathematics for Engineers (Dept. of Mathematics)
2017-20	<i>Small group teaching:</i> Calculus in Several Dimensions, Applied Mathematics and Newtonian Mechanics, Mathematical Methods, Bridging Course, Year 12 Summer Challenge, Quantitative Methods (Dept. of Mathematics)

Research supervision

POSTDOCTORAL SUPERVISION

2024-25	Michael Mahony. "TIMELY - Validating an up-to-date dataset of internal migration to inform local and regional-scale spatial planning strategies in Chile". <i>University of Liverpool</i> .
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PHD SUPERVISION

- 2024- Emiliano Beltrán Alipio. “Safeguarding urban communities: Using data science to determine the impact of the spatial structure of urban activities and residential segregation on the spreading of infectious diseases”. First co-supervisor, *U. of Liverpool*.
- 2023- Alice Varley. “Understanding social and spatial inequalities in common mental health disorders”. First supervisor, *U. of Liverpool*.
- 2022-26 Andrea Nasuto. “Understanding the role of social media in promoting anti-migration sentiment and hate crime”. Second supervisor, *U. of Liverpool*.
- 2022-24 Niall Newsham. “Understanding the extent, demographic causes, consequences and future of population decline in Europe”. Second supervisor, *U. of Liverpool*.

RESEARCH ASSISTANT SUPERVISION

- 2025-26 Emiliano Beltrán Alipio. “Building Imago-HASP collaborative infrastructure for joint data products linking satellite, synthetic and trajectory data”. *University of Liverpool*.
- 2025-26 Rodgers Iradukunda. “Building Imago-HASP collaborative infrastructure for joint data products linking satellite, synthetic and trajectory data”. *University of Liverpool*.
- 2025-26 Leran Zheng. “Building Imago-HASP collaborative infrastructure for joint data products linking satellite, synthetic and trajectory data”. *University of Leeds*.
- 2024-25 Giada Garofani. “Establishing a new cross-sector, collaborative agenda for data-driven transport planning in Liverpool City Region”. *University of Liverpool*.
- 2022-23 Andrea Nasuto. “RECAST – Strengthening policy collaboration with the United Nations: Recovery and resilience through digital footprint data”. *University of Liverpool*.
- 2022-23 Ruth Neville. “RECAST – Strengthening policy collaboration with the United Nations: Recovery and resilience through digital footprint data”. *University of Liverpool*.

Service

POLICY AND ADVISORY ROLES

- 2025- Member of UN Committee of Experts on Big Data and Data Science for Official Statistics – Task Team on Mobile Phone Data, Transport Group – <https://unstats.un.org/bigdata/>
- 2024 “LCRCA Pedestrian Infrastructure, Accessibility & Safety”. Liverpool City Region Combined Authority – <https://www.liverpoolcityregion-ca.gov.uk>

EDITORIAL ROLES

- 2025- Editorial Board, Annals of the American Association of Geographers
- 2025- Associate Editor, Applied Network Science
- 2023- Junior Editor, Computational Urban Science
- 2017-21 Editor, [Chalkdust](#)

PHD EXTERNAL EXAMINER

- 2024 Irene Meta. “City Physiology Framework: A Conceptual Approach to Improve Urban Simulations”. *Universitat Internacional de Catalunya, Spain*.
- 2023 Jonathan Kent. “Urban Fabric Designed for Diversity: How the Structure of Great European Cities Influences Attitudes and Behavior”. *Universitat Pompeu Fabra, Spain*.

JOURNAL REVIEWS

Nature Cities · Nature Communications · International Journal of Geographic Information Science · Journal of Computational Urban Science · Cities · Environment and Planning B: Urban Analytics and City Science · Geographical Analysis · Computational Urban Science · Heliyon · Journal of Statistical Physics · PLOS ONE · Sage Research Methods

GRANT REVIEWS

2024- Caixa – Social Research Call

CONFERENCE ROLES

2026- Programme Chair. Complenet 2026, Zaragoza, Spain.
2024 Scientific committee. Netsci 2024. International School and Conference on Network Science, Québec City, Canada.

LEADERSHIP AND EQUALITY & DIVERSITY

2025- Faculty Chair of Early Career Academic Leadership Group, Faculty of Science and Engineering, University of Liverpool
2023-25 Early Career Academic Lead Representative, School of Environmental Sciences, University of Liverpool
2018-21 Equality & Diversity Committee PhD representative, Department of Mathematics, UCL
· Contributed to the successful application for the Athena SWAN Silver Award 2021

OUTREACH

2024 Workshop for Year 12 students: “Mapping connections: a journey through Networks and Geography”. University of Liverpool, UK
2019 Drop-in activity at IMA Festival of Mathematics and its Applications, University of Greenwich, UK
2019 Talk: Understanding Complexity, Girls in Maths Day, St John Bosco College, London, UK

Relevant skills

Python and R (10+ years experience): data analysis, mathematical and statistical modelling, geospatial analysis, machine learning, simulations, optimisation, visualisation

Other programming languages: Matlab, C++/C, FORTRAN

Experience working with data at the **Secure Research Service of the UK Office for National Statistics**, including being **trained as a Full Accredited Researcher**

Software and cloud-based services: GitHub, LaTeX, Overleaf, Quarto, Microsoft Office, etc.

Languages: English (fluent), Spanish (native), Catalan (native), French (intermediate)

Last updated: March 2, 2026